

AssetTrace Node

Project Presentation

AssetTrace Node is a local system designed for machine park management, fault memory, and maintenance pattern analysis.

It was developed as an operational development mode of a broader mother architecture, but it has its own clear identity: it is not just a chatbot, not just a database, and not a closed predictive system. It is a local node that collects technical information, organizes machine events, connects faults to assets, and helps recurring signals emerge over time.

The core idea is simple: every machine has a history.

A 3D printer, a Raspberry Pi, a local server, a sensor, or any technical system is not an isolated object. It produces events, anomalies, states, faults, repairs, and modifications. When this information is scattered, every problem appears new. When it is collected and organized, recurring patterns begin to appear.

AssetTrace Node was created to transform technical faults and machine events into structured memory.

The system allows the user to register machines, update their status, add technical events, store faults, connect external observations, and analyze recurring patterns. Each machine can have a history, a criticality level, a health index, and an evolutionary trace. In this way, the machine park is not managed as a simple list of devices, but as a living technical system observed over time.

One of the central components is the **patterns module**. This module does not perform heavy machine learning and does not modify the AI model. Instead, it works on the local data already collected: events, faults, machine states, recurring terms, and criticality levels. From these elements it generates a health index, highlights fragile machines, and suggests the next minimal action.

AssetTrace Node also integrates with Ollama, allowing it to use a local language model for reasoning over the collected data. However, the project follows an important principle: the system does not make final decisions and does not act directly on the physical world. Responsibility remains with the user. AssetTrace observes, organizes, highlights, and proposes.

Another important component is governed web search. The web module can be used to look for updated information, technical documentation, or similar cases online. However, web results are not treated as automatic truth. They are external signals that must be evaluated according to source, freshness, reliability, contradiction, and risk.

AssetTrace Node is therefore a first step toward a local maintenance intelligence system: lightweight, extensible, modular, and understandable. It can initially be used from the

terminal, but its structure is designed to evolve toward dashboards, automatic reports, OctoPrint integration, semantic memory, and more advanced predictive maintenance indicators.

In summary, AssetTrace Node is a local maintenance intelligence node.

It does not replace the technician.

It helps the technician preserve memory, recognize recurring patterns, and make more structured maintenance decisions.